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SESAME: The Los Alamos National Laboratory's Tabulated Equation of State Database Description with Extensions for Multi-phase Representations

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This document describes details of the structure and content of the SESAME database. This includes details on both legacy table structures and new tables structures, which improve representations of phase specific thermodynamic properties and explicitly provide tabulated phase boundaries. Such database augmentations will provide improved physical representations of equilibrium mixed phase regions. In addition to accurate modeling of mixed phases, the new phase specific tables provide a framework for the implementation of important non-equilibrium kinetic modeling of phase transitions.

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I. INTRODUCTION

Modeling of the thermodynamic equation of state (EOS) of various materials has had a long storied tradition at LANL. As early as 1949 Feynman, Metropolis, and Teller published a paper presenting EOS values for some elements and a methodology for calculating the EOS at high compression [1]. Cowan and Ashkin made notable methodology improvements for compressed materials throughout the 1950’s and beyond [2]. In 1971 Jack Barnes and Jerry Rood created the SESAME database and by 1972 the database became publicly available. This database was designed around materials identified by a material ID number. The thermodynamic state variables are then tabulated on a density (ρ) and temperature (T) grid. The original modeling centered around segregating the EOS into a three part decomposition of the Helmholtz free energy [3],

$$A(\rho, T) = A^c(\rho) + A^{ion}(\rho, T) + A^{ele}(\rho, T) \quad (1)$$

where $A^c(\rho) = A(\rho, T=0)$, known as the cold curve, models the compression response of the material as a function of density. A^{ion} captures the energy contributions due to ionic/atomic vibrations, rotations, and translations. Finally A^{ele} models the response to electronic excitations. This decomposition is evident in the structure and design of SESAME and can clearly be seen in Table II, presented later, where the total EOS corresponds to the 301 table. Additionally, the cold curve, ion EOS and electronic EOS are tabulated in the 306, 305, and 304 tables respectively.

The SESAME database is celebrating its 53rd anniversary of its public release this year, and as with any database this old, over the years it has become necessary to rework the details. This includes expanding the definition of table types and the scope of data represented to reflect advances in materials physics and code advancements, which make modeling physics details possible. The last document that comprehensively detailed the SESAME structure was locally published in 1992 [4]. This report will regurgitate some information from that report, modifying where required, adding long existing tables that were not included, and defining brand new tables which will facilitate the future of explicit phase-aware material properties. This extension to a multi-phase material description requires tabulating phases individually, tabulation of phase boundaries, triple point tables, and naturally leads to extension of phase specific shear. Phase specific “grey” (or “gray”) opacity and conductivity tables will not be expanded at this point but it would be straight forward to use the generalized frame work to do so in the future when required.

The inclusion and formal definition of phase-specific tables in SESAME is long overdue. As host-codes required access to individual phases in addition to a flattened total EOS, EOS developers quickly adapted by creating multiple materials, one flattened EOS with a formal SESAME material ID number and a separate material with a “unique” material ID number for each phase. Though effective and quick, this has lead to an explosion of SESAME materials with no formal mapping between a given flattened EOS material and individual phases that were used to create it. By including everything under the definition of one material going forward, we ensure a self-consistent pedigree for the entire material.

This report will include a section on physics notation and nomenclature (Section II), discussion of existing and new tables and their structure (Section III), data requirements and expectations for the new phase-boundary tables (Section IV K), a short section and reference to the newly released ASCII2 SESAME file format (Section V), and an overview of existing software made to directly interact with SESAME data.

II. NOTATION

As for any scientific field, clear definitions for symbols and conventions are important. Table I summarizes a common notation used for this document. We have tried to be consistent with the notation adopted in the EOSPAC6 manual[5], the EOS project’s recommend method for interacting with SESAME data. That being said there are notable differences in notation used between various EOS production codes, reports, and user manuals. For example, OPENSESAMe[6], a code used to create SESAME tables uses E instead of U for internal energy. It is also common for EOS literature

to use F (a physics convention) instead of A (a chemistry convention) when describing the Helmholtz free-energy. In general, the EOS program at LANL has commonly used the term free-energy when referring to Helmholtz free-energy. The distinction is now important because of the addition of tabulated values of Gibbs free-energy[7] into some of the newly defined table types.

TABLE I: Notation, units, and descriptions of physics notation used in this document.

Symbol	Units	Description
ρ	Mg/m ³ , (g/cm ³)	Density (units are formally equivalent)
V	m ³ /Mg	Specific volume
T	K	Temperature
P	GPa	Pressure
U	MJ/kg	Internal energy
H	MJ/kg	Enthalpy
S	MJ/kg/K	Entropy
A	MJ/kg	Helmholtz free-energy
G	MJ/kg	Gibbs free-energy
$\mathbb{G}$	GPa	Shear modulus
B_T	GPa	Isothermal bulk modulus
B_S	GPa	Adiabatic bulk modulus
C_V	MJ/kg/K	Constant volume heat capacity
C_P	MJ/kg/K	Constant pressure heat capacity
α_P	K ⁻¹	Volumetric thermal expansion at constant pressure
Γ		Grüneisen parameter
ϕ		Phase number
λ		Phase fraction [0-1]

III. LEGACY AND PHASE-SPECIFIC SESAME TABLES

Legacy tables (Table II), in addition to new tables (Table III) are included in this section for convenience/completeness. SESAME has grown and evolved over many years, and the 1992 SESAME document[4], at the time of its publication, was a complete description of all of the tables in circulation. The purpose of this document is to provide additional details in the form of a modern, comprehensive description of all of the tables currently in circulation. A notable addition to the current collection of tables is the 321 mass fraction table which was included as a first pass at a multi-phase EOS. A not-so notable addition is a definition of the now obsolete 432 table[8] and the expansion of the 431 definition[9]. New table types extend on this work and it is hoped that the future will allow the deprecation of the 321 table for all new materials. A particular challenge for the EOS project is moving forward with better physical representations while maintaining compatibility with legacy data set still in production. These two goals

are not always mutually exclusive, and they often impose significant design hurdles. That being said, attempts have been made to retain as much of the current conventions for table structures, and these new, phase-specific tables should allow for improved interpolation behavior and physical representation of phase changes which were previously troublesome or smoothed for numerical stability.

TABLE II: Legacy SESAME data records.

Table number	Description
100	JSON strings for user defined tables[10]
101-199	Comments
201	Mean Atomic Number, Mean Atomic Mass, Normal (solid) Density, Solid Bulk Modulus, Exchange coefficient
301	Total EOS (304 + 305 + 306)
303	Ion EOS plus Cold Curve (305 + 306)
304	Electron EOS
305	Ion EOS (including zero point correction)
306	Cold Curve (No zero point correction)
311	Total EOS (Maxwell constructed)
321	Mass Fractions for <i>flattened</i> Total EOS
401	Vaporization Table
411	Solid Melt Table
412	Liquid Melt Table
431	Shear Modulus Table (<i>GPa</i>) [9]
432	Cold Shear Modulus and Shear at melt Table (Obsolete)[8]
501	Opacity Grid Boundary: Calculated vs. Interpolated
502	Rosseland Mean Opacity ($\text{cm}^2 \text{g}^{-1}$)
503	Electron Conductive Opacity ($\text{cm}^2 \text{g}^{-1}$), Hubbard-Lampe Opacity Model
504	Mean Ion Charge (free electrons per atom), Hubbard-Lampe Opacity Model
505	Planck Mean Opacity ($\text{cm}^2 \text{g}^{-1}$)
601	Mean Ion Charge (free electrons per atom), Ziman Conductivity Model
602	Electrical Conductivity (sec^{-1})
603	Thermal Conductivity ($\text{cm}^{-1} \text{sec}^{-1}$)
604	Thermoelectric Coefficient ($\text{cm}^{-1} \text{sec}^{-1}$)
605	Electron Conductive Opacity ($\text{cm}^2 \text{g}^{-1}$), Ziman Conductivity Model

TABLE III: Phase-Specific SESAME data records.

Table number	Description
<i>101xx ...</i>	<i>Modeled after current table numbering (Table II) where ‘xx’ indicates the phase number</i>
10101-10199	Phase specific comment tables
20101-20199	Phase specific Mean Atomic Number, Mean Atomic Mass, Normal (solid) Density, Solid Bulk Modulus, Exchange coefficient
30101-30199	Phase specific Total EOS (304xx + 305xx + 306xx)
30301-30399	Phase specific Ion EOS plus Cold Curve (305xx + 306xx)
30401-30499	Phase specific Electron EOS
30501-30599	Phase specific Ion EOS (including zero point correction)
30601-30699	Phase specific Cold Curve (No zero point correction)
43101-43199	Phase specific Cold Shear Modulus
<i>421yyxx</i>	<i>Lower phase boundary; ‘yy’ indicates property tabulated where ‘xx’ indicates the phase number</i>
4210101-4210199	Temperature along lower phase boundary, T (K)
4210201-4210299	Pressure along lower phase boundary, P (GPa)
4210301-4210399	Gibbs free-energy along lower phase boundary, G (MJ/kg)
4210401-4210499	Density along lower phase boundary, ρ (Mg/m ³)
4210501-4210599	Internal energy along lower phase boundary, U (MJ/kg)
4210601-4210699	Helmholtz free-energy along lower phase boundary, A (MJ/kg)
4210701-4210799	Phase adjacent to lower phase boundary, ϕ_{aj}
4210801-4210899	Heat capacity at constant volume along lower phase boundary, C_V (MJ/kg/K)
4210901-4210999	Heat capacity at constant pressure along lower phase boundary, C_P (MJ/kg/K)
4211001-4211099	Adiabatic bulk modulus along lower phase boundary, B_S (GPa)
4211101-4211199	Isothermal bulk modulus along lower phase boundary, B_T (GPa)
4211201-4211299	Thermal expansion coefficient along lower phase boundary, α_P (K ⁻¹)
4211301-4211399	Grüneisen parameter along lower phase boundary, Γ
<i>422yyxx</i>	<i>Upper phase boundary; ‘yy’ indicates property tabulated where ‘xx’ indicates the phase number</i>
4220101-4220199	Temperature along upper phase boundary, T (K)
4220201-4220299	Pressure along upper phase boundary, P (GPa)
4220301-4220399	Gibbs free-energy along upper phase boundary, G (MJ/kg)
4220401-4220499	Density along upper phase boundary, ρ (Mg/m ³)
4220501-4220599	Internal energy along upper phase boundary, U (MJ/kg)

Continued on next page

TABLE III: Phase-Specific SESAME data records. (Continued from previous page.)

Table number	Description
4210601-4220699	Helmholtz free-energy along upper phase boundary, A (MJ/kg)
4220701-4220799	Phase adjacent to upper phase boundary, ϕ_{aj}
4220801-4220899	Heat capacity at constant volume along upper phase boundary, C_V ($MJ/kg/K$)
4220901-4220999	Heat capacity at constant pressure along upper phase boundary, C_P ($MJ/kg/K$)
4221001-4221099	Adiabatic bulk modulus along upper phase boundary, B_S (GPa)
4221101-4221199	Isothermal bulk modulus along upper phase boundary, B_T (GPa)
4221201-4221299	Thermal expansion coefficient upper lower phase boundary, α_P (K^{-1})
4221301-4221399	Grüneisen parameter along upper phase boundary, Γ
423zz	Single triple point table where ‘zz’ is an iterator to identify a specific triple point
42301-42399	Triple point table

A. Structure of SESAME Data

The data for each table in SESAME is stored as a 1-D array (single stream of numbers). Typically, the first couple of values in the array determine the extents of the arrays for the independent variables. The two-dimensional tables in SESAME are tabulated as functions of density and temperature, $f(\rho, T)$. There are also one-dimensional tables tabulated as functions of density, $f(\rho)$. Tables IV to VIII list the data structures of each SESAME construct, the column *Index*, also often referred to as *WORD*, is a relative index offset where each new array begins. It should be noted that this is only in reference to the order of the data and includes neither an ascii-file-format-specific header nor a binary-file-format-specific index that identifies which SESAME table is being tabulated. These format specific headers are described in Section V.

TABLE IV: Detail on table data structure for 200 and 300 style tables.

Table 201, 20101-20199		
Index	Variable	Description
1	ZBAR	Mean atomic number
2	ABAR	Mean atomic mass (g/mol)
3	ρ_0	Normal (solid) density (Mg/m^3)
4	B_S	Adiabatic bulk modulus at ρ_0 (GPa)
5	CEX	Exchange coefficient
Table 301-305, 311, 30101-30599 i=1,NR; j=1,NT		
Index	Variable	Description

Continued on next page

TABLE IV: Detail on table data structure for 200 and 300 style tables. *(Continued from previous page.)*

1	NR	Number of densities
2	NT	Number of temperatures
3	ρ_i	Density array (Mg/m ³)
NR+3	T_j	Temperature array (K)
NR+NT+3	P_{ij}	Pressure array (GPa)
(NR*NT+NR+NT+3	U_{ij}	Internal-energy array (MJ/kg)
2(NR*NT+NR+NT+3	A_{ij}	Helmholtz free-energy (MJ/kg)[11]
Table 306, 30601-30699	i=1,NR	
Index	Variable	Description
1	NR	Number of densities
2	NT(=1)	Number of temperatures is singular
3	ρ_i	Density array (Mg/m ³)
NR+3	$T(=0)$	Scalar temperature value is zero (K)
NR+4	P_i	Pressure array (GPa)
2NR+4	U_i	Internal-energy array (MJ/kg)
3NR+4	A_i	Helmholtz free-energy (MJ/kg)[11]
Table 321	i=1,NR; j=1,NT	
Index	Variable	Description
1	NR	Number of densities
2	NT	Number of temperatures
3	ρ_i	Density array (Mg/m ³)
NR+3	T_j	Temperature array (K)
NR+NT+3	λ_{ij}	Mass fraction of phase 1
(NR*NT)+NR+NT+3	λ_{ij}	Mass fraction of phase 2
2(NR*NT)+NR+NT+3	λ_{ij}	Mass fraction of phase 3
$\vdots$	$\vdots$	
(N-1)(NR*NT)+NR+NT+3	λ_{ij}	Mass fraction of phase N, where $N = \frac{(NWORDS-2-NR-NT)}{(NR \times NT)}$ and $NWORDS$ is the total size of the 321 table

TABLE V: Detail on table data structure for vapor dome and melt tables.

Table 401 i=1,NT

Index	Variable	Description
1	NT	Number of temperatures
2	P_i	Vapor pressure array (GPa)
NT+2	T_i	Temperature array (K)
2NT+2	ρ_i^g	Vapor density array (Mg/m ³)
3NT+2	ρ_i^l	Liquid density array (Mg/m ³)
4NT+2	U_i^g	Vapor internal-energy array (MJ/kg)
5NT+2	U_i^l	Liquid internal-energy array (MJ/kg)
6NT+2	A_i^g	Vapor Helmholtz free-energy (MJ/kg)
7NT+2	A_i^l	Liquid Helmholtz free-energy (MJ/kg)

Table 411 i=1,NR

Index	Variable	Description
1	NR	Number of densities
2	NT(=1)	Number of temperatures is singular
3	ρ_i^s	Density array of solid at melt (Mg/m ³)
NR+3	$T(=0)$	Scalar temperature value is zero (K)
NR+4	T_i^m	Melt temperature (K)
2NR+4	P_i^m	Melt pressure (GPa)
3NR+4	U_i^s	Internal-energy of solid at melt (MJ/kg)
4NR+4	A_i^s	Helmholtz free-energy of solid at melt (MJ/kg)

Table 412 i=1,NR

Index	Variable	Description
1	NR	Number of densities
2	NT(=1)	Number of temperatures is singular
3	ρ_i^l	Density array of liquid at melt (Mg/m ³)
NR+3	$T(=0)$	Scalar temperature value is zero (K)
NR+4	T_i^m	Melt temperature (K)
2NR+4	P_i^m	Melt pressure (GPa)
3NR+4	U_i^l	Internal-energy of liquid at melt (MJ/kg)
4NR+4	A_i^l	Helmholtz free-energy of liquid at melt (MJ/kg)

TABLE VI: Detail on table data structure for shear and opacity tables.

Table 431, 43101-43199 $i=1, NR$		
Index	Variable	Description
1	NR	Number of densities
2	NT(=1)	Number of temperatures is singular
3	ρ_i	Density array (Mg/m ³)
NR+3	$T(=0)$	Scalar temperature value is zero (K)
NR+4	$\mathbb{G}_i$	Cold Shear modulus array (GPa)
2*NR+4	$\mathbb{G}_i$	Shear modulus array at melt (GPa)[9]
Table 501 $i=1, N$		
Index	Variable	Description
1	N	Number of temperature and densities pairs
2	T_i, ρ_i	$\log_{10} T_i, \log_{10} \rho_i$
Table 502-506, 601-605 $i=1, NR; j=1, NT$		
Index	Variable	Description
1	NR	Number of densities
2	NT	Number of temperatures
3	ρ_i	$\log_{10}$ Density array (g/cm ³)
NR+3	T_i	$\log_{10}$ Temperature array (eV)
NR+NT+3+4	X_i	$\log_{10}$ quantity from Table II

TABLE VII: Detail on table data structure for triple-point data. For a triple-point the G, T , and P are common according to eq. (4) they are only tabulated once followed by triplets of thermodynamic data corresponding to each phase.

Table 42301-42399 Triple-point		
where i, j , and k identify the individual phases		
Index	Variable	Description
1	T	Temperature (K)
2	P	Pressure (GPa)
3	G	Gibbs free-energy (MJ/kg)
4	ϕ_i, ϕ_j, ϕ_k	Phase identifiers: i, j, k (integers)
7	ρ_i, ρ_j, ρ_k	Density of phases i, j , and k (Mg/m ³)
10	U_i, U_j, U_k	Internal energy of phases i, j , and k (MJ/kg)

Continued on next page

TABLE VII: Detail on table data structure for triple-point data. For a triple-point the G, T , and P are common according to eq. (4) they are only tabulated once followed by triplets of thermodynamic data corresponding to each phase. *(Continued from previous page.)*

13	A_i, A_j, A_k	Helmholtz free-energy of phases i, j, and k (MJ/kg)
16	C_{Vi}, C_{Vj}, C_{Vk}	Constant volume heat capacity of phases i, j, and k (MJ/kg/K)
19	C_{Pi}, C_{Pj}, C_{Pk}	Constant pressure heat capacity of phases i, j, and k (MJ/kg/K)
22	B_{Si}, B_{Sj}, B_{Sk}	Adiabatic bulk modulus of phases i, j, and k (GPa)
25	B_{Ti}, B_{Tj}, B_{Tk}	Isothermal bulk modulus of phases i, j, and k (GPa)
28	$\alpha_{Pi}, \alpha_{Pj}, \alpha_{Pk}$	Isobaric thermal expansion of phases i, j, and k (1/K)
31	$\Gamma_i, \Gamma_j, \Gamma_k$	Grüneisen parameter of phases i, j, and k

TABLE VIII: Detail on table data structure for phase boundary tables, where yy (i.e., valid values are 01-14) is property index, xx is the phase number (i.e., valid values are 01-99). All of these numbered tables employ a common table structure; however, the value of NR will only be consistent for a given phase number, xx, and it is typically different for corresponding arrays in 421yyxx and 422yyxx.

Common Table Structure			
	Index	Variable	Description
	1	NR	number of values
	2	$symbol(yy)$	$description$
Table 421yyxx	i=1,NR	Lower boundary of phase xx	
Table Number	Variable(yy)	Description	
42101xx	T_i	Temperature array (K)	
42102xx	P_i	Pressure array (GPa)	
42103xx	G_i	Gibbs free-energy array (MJ/kg/K)	
42104xx	ρ_i	Density array (Mg/m ³)	
42105xx	U_i	Internal energy array (MJ/kg/K)	
42106xx	A_i	Helmholtz free-energy array (MJ/kg/K)	
42107xx	ϕ_i^{aj}	Adjacent phase identifier array (integer)	
42108xx	C_{Vi}	Constant volume heat capacity array (MJ/kg/K)	
42109xx	C_{Pi}	Constant pressure heat capacity array (MJ/kg/K)	
42110xx	B_{Si}	Adiabatic bulk modulus array (GPa)	
42111xx	B_{Ti}	Isothermal bulk modulus array (GPa)	
42112xx	α_{Pi}	Isobaric thermal expansion array (1/K)	
42113xx	Γ_i	Grüneisen parameter array	

Continued on next page

TABLE VIII: Detail on table data structure for phase boundary tables, where yy (i.e., valid values are 01-14) is property index, xx is the phase number (i.e., valid values are 01-99). All of these numbered tables employ a common table structure; however, the value of NR will only be consistent for a given phase number, xx, and it is typically different for corresponding arrays in 421yyxx and 422yyxx. (Continued from previous page.)

Table 422yyxx		
i=1,NR Upper boundary of phase xx		
Table Number	Variable(yy)	Description
42201xx	T_i	Temperature array (K)
42202xx	P_i	Pressure array (GPa)
42203xx	G_i	Gibbs free-energy array (MJ/kg/K)
42204xx	ρ_i	Density array (Mg/m ³)
42205xx	U_i	Internal energy array (MJ/kg/K)
42206xx	A_i	Helmholtz free-energy array (MJ/kg/K)
42207xx	ϕ_i^{aj}	Adjacent phase identifier array (integer)
42208xx	C_{Vi}	Constant volume heat capacity array (MJ/kg/K)
42209xx	C_{Pi}	Constant pressure heat capacity array (MJ/kg/K)
42210xx	B_{Si}	Adiabatic bulk modulus array (GPa)
42211xx	B_{Ti}	Isothermal bulk modulus array (GPa)
42212xx	α_{Pi}	Isobaric thermal expansion array (1/K)
42213xx	Γ_i	Grüneisen parameter array

IV. GENERAL TABLE DISCUSSIONS

This section provides some general descriptions and/or discussions about various groups of tables, and the data they contain, as defined for and exists within the SESAME database.

A. 101-style Comment Tables

The 101 comment table is designed to contain specific metadata for a SESAME material, and the possible table number values (i.e., a **Table_ID**) are 101 and the range 10101-10199. Tables 10101-10199 are intended to carry the same specific metadata relevant to each phase. The phase specific metadata for phase 1 is contained in table 10101, for phase 2 in 10102, etcetera. Keywords in the 101-style table are described in the following bulleted list (bold; including period, '.'). Note that **Table_ID** 101 and 10101-10199 are specifically reserved for 101 style data. Although it has not strictly enforced throughout the history of LANL's[12] SESAME parsers, certain conventions are assumed by LANL EOS software and should be followed. The case of a keyword is not enforced (i.e., case-insensitive).

- **material.** Material or phase name (z=atomic number, a=atomic weight)
- **source.** Author, organization detail.
- **date.** Date the material was created.

- **refs.** Published references.
- **comp.** Composition of the material.
- **code.** Named software used to create the material data.
- **classification.** Unclassified, CUI::SP-EXPT (formally Export Controlled), Secret Restricted Data or other LANL approved classification marking[13].
- **phases.** An optional string containing a comma-delimited list of the number of phases followed by an ordered phase names. Some conventions are generally assumed for the string content:
 - It is assumed that any leading and/or trailing white space will be ignored for each phase name.
 - The left-to-right order of these phase names must correspond to the data order in any other table that contains phase-specific data (e.g., 321, 10101, 10102, 10103, 30101, 30102, 30103, etc.).
 - The first phase listed corresponds to the ambient phase data.
 - The last phase listed is the high-temperature phase, which usually corresponds to the liquid phase data.

Some fabricated examples of the keyword-value pairs are shown in (Table IX).

TABLE IX: 101 comment table keywords and examples used in this document.

Keyword	Example
material.	material. Aluminum (z=13.0, a=26.9815)/
source.	source. J. Edgar Hoover, Zx-10/
date.	date. Aug. 28, 1998/
refs.	refs. LAUR-99-9090/
comp.	comp. Al /
codes.	codes. Foxtrot Express (ver. 1.40a)/
classification.	classification. Unclassified/
phases.	phases. 3, bcc, hcp, liquid /

B. Generic comment table

Generic comment tables are potentially-numbered with a `Table_ID` in the range of 102-199, and they have neither any explicit keywords definitions nor any required organizational structure. These comments are intended to provide the consumer of SESAME data with useful information about the data pedigree and history.

C. Material constants: 201

Several scalar material constants are stored in the 201 table (table IV). The mean atomic number, mean atomic mass, and normal density are consistently tabulated, and they correspond to the values supplied via the OPENSESAME[6] input; however, a large majority of the adiabatic bulk modulus and exchange coefficient data are assigned useless zero values within the aggregated SESAME database.

D. 300 series tables

The 300 series tables represent SESAME’s fundamental EOS data. As alluded in section III A and implied in table IV, the 300 series tables contain five main arrays. There exist two one-dimensional independent variable arrays containing density and temperature data, and there are three two-dimensional, column-major [14] arrays containing density- and temperature-dependent thermodynamic functions: pressure, internal energy, and Helmholtz free energy. The 306 style table also follows this structure, but the temperature array only contains one value, $T = 0K$. What the table IV fails to convey is the implied physical and numerical requirements of the data stored in 300 series tables. The following is an attempt to explicitly list the implicit requirements of these data.

- The density array shall be monotonically increasing with a minimum possible value of zero.
- The temperature array shall be monotonically increasing with a minimum possible value of zero.
- Best practice is for the temperature array to start with zero.
- Internal-energy shall monotonically increase as a function of temperature at a given density.
- Helmholtz free-energy shall monotonically decrease as a function of temperature at a given density.
- Helmholtz free-energy and internal-energy shall be identical at $T = 0$.
- The dependent data (i.e., pressure, internal energy, and Helmholtz free energy) in the 301 should equal the sum of the data in each of the 304, 305, and 306 tables.
- Density and temperature arrays should be approximately evenly log-spaced, with extra resolution where the absolute values of the first and/or second derivatives are large.
- Other thermodynamic consistency checks, based on Maxwell equations, shall be enforced.

E. Mass fraction table: 321

The 321 table is a variation of a 300-series table that contains equilibrium mass fraction data for each material phase corresponding to the tabulated density and temperature grid of an EOS. The data associated with the liquid (i.e., fluid) phase of the EOS is tabulated last; In other words, the liquid phase index will be the largest of all tabulated phases within a 321 table (e.g., phase 1: bcc, phase 2: hcp, phase 3: liquid). The 321 table is similar to a 301 table in that there are multiple two-dimensional, density- and temperature-dependent arrays stored in sequence; however, the number of these arrays is not limited to a maximum of three as is the case for the 301 table. Instead, the number of material phases that are represented is what determines the total array count. Table IV contains a visual representation of how data is organized in a 321 table, directly contrasted to how data in a 301 table is organized. As described in the previous section (section IV D), the following is an attempt to explicitly list the implicit requirements of these mass fraction data.

- The density array shall be monotonically increasing with a minimum possible value of zero, and correspond to the related 300-series tables.
- The temperature array shall be monotonically increasing with a minimum possible value of zero, and correspond to the related 300-series tables.
- Best practice is for the temperature array to start with zero.
- The dependent data (i.e., equilibrium mass fraction data) in all of the individual phase-specific arrays should sum to a value of 1 at each (density,temperature) grid location.
- Density and temperature arrays should be approximately evenly log-spaced as is done for the 300-series tables described in section IV D.

F. Vapor-dome table: 401

The 401 table is designed to contain a one-dimensional grid representation of the EOS' vapor dome phase boundary. This boundary is represented by tabulated data for the following thermodynamic quantities: Temperature (K), Vapor Pressure (GPa), Vapor Density on coexistence line (Mg/m^3), Liquid-or-Solid Density on coexistence line (Mg/m^3), Vapor Specific-Internal-Energy (MJ/kg), Liquid-or-Solid Specific-Internal-Energy (MJ/kg), Vapor Specific-Helmholtz-Free-Energy (MJ/kg), and Liquid-or-Solid Specific-Helmholtz-Free-Energy (MJ/kg). The Vapor-specific data represent the vapor dome phase boundary for densities less than or equal to the EOS' critical point. Alternatively, the Liquid-or-Solid-specific data represent the vapor dome phase boundary for densities greater than or equal to the EOS' critical point. Both the pressure and temperature arrays are common to the other Vapor- and Liquid-or-Solid-specific arrays stored within the 401 table. In other words, either the temperature or pressure arrays can be used as a common grid for the other tabulated thermodynamic quantities. Table V contains a visual representation of how data is organized in a 401 table.

G. Melt tables: 411 and 412

The 411 and 412 tables contain a tabulated representation of the EOS solid-to-liquid melt transition region. This phase change region is bounded by both a solid (411) and a liquid (412) tabulated boundaries that are density-dependent arrays of the following thermodynamic data: Melt Temperature (K), Melt Pressure (GPa), Melt Specific-Internal-Energy (MJ/kg), and Melt Specific-Helmholtz-Free-Energy (MJ/kg). Table V contains a visual representation of how data is organized in both a 411 and a 412 tables. The density grid in these tables should be monotonically increasing. Ideally 411 and 412 tables should be consistent with the 301 table for a given material ID. For example, starting with a tabulated density from the 411 table and retrieving the corresponding temperature and pressure tabulated at that density point, one should be able to interpolate a consistent pressure from the 301 table at the same density and temperature. In practice, this is only true for 301 tables constructed where an explicit melt transition was modeled in the 301 table, 411 and 412 tables generated alongside the multiphase-flattening algorithm were introduced to the database around 2010. For legacy materials and materials using legacy modeling techniques most of melt tables were generated so that the 411 table is consistent with the 301 and the 412 table is shifted for a fixed change in entropy, typically not consistent with the 301 table.

H. Shear: the expansion of the 431 table and the demise of the 432 table

The 431 table contains cold shear modulus (GPa) and optionally a second array containing the shear modulus at the melting temperature as a function of density. Cold shear modulus in this context is the shear modulus at $T = 0K$. When using the shear modulus at melt array, the melting temperature at a given density is not tabulated explicitly so one should use these in conjunction with the melt tables described in Section IV G. Table VI contains a visual representation of how data is organized. This new extension of the 431 to optionally include shear modulus at melt is an adaptation intended to include data that was previously stored in the undocumented 432 table. The now obsolete 432 table structure is defined exactly like the expanded 431 table where the second array containing shear at melt data is not optional. In current SESAME releases these tables have been combined into a single 431 table. This document also defines phase-specific 431-style tables 43101-43199 as place holders for future materials that might take advantage of phase dependent shear modulus.

I. Gray opacity tables: 500 series

The 500 series tables contain the SESAME opacity data, which are so-called “gray” (or “grey”) Rosseland and Planck mean opacities, and they are provided for a prescribed grid of temperatures and densities. The Rosseland and Planck mean data are contained within the 502 and 505 SESAME tables respectively. (For completeness, we mention that the 504 table contains the mean ion charge (i.e., free electrons per atom) that is consistent with the opacity calculations. Also, the 503 table contains the electron conductive opacity, when available[15]. These SESAME opacity data are obtained by processing the Los Alamos OPLIB opacity database[16], which contains detailed, frequency-dependent opacities for pure elements. The fundamental atomic data that are needed to calculate the OPLIB opacities are generated with the Los Alamos suite of atomic physics codes[17]. The SESAME gray opacity data are obtained by integrating the frequency-dependent OPLIB data, convolved with the appropriate (Rosseland or Planck) weighting function. This processing of the OPLIB opacities is done with the TOPS codes[18, 19], which can also be used in a more general context to produce opacity files, for both pure elements and mixtures, that can be used in radiation-hydrodynamics codes.

As seen in table VI, the 500 series tables store logarithmically-transformed (base 10) data.

J. Conductivity tables: 600 series

The 600 series tables contain electrical conductivity (602), thermal conductivity (603), the thermoelectric coefficient (604) and the electron conductive opacity (605). These are tabulated in Gaussian units. Table 601 reports the mean ion charge, which, in older Ziman-like conductivity models, is a key parameter. For modern tables this quantity is often not as directly related to the transport quantities. Recent 600 tables have been produced by direct calculation, using density functional theory and kinetic theory [20, 21]. Older tables used the Ziman model and should be used with caution [20, 22].

As specified in table VI, the 600 series tables contain logarithmically-transformed (base 10) data.

K. Phase-boundary tables, 421yyxx and 422yyxx series

The equilibrium phase boundary between two phases can be found using either the Helmholtz free-energy and a lever rule or by equating the Gibbs free-energy of two phases i and j .

$$G_i(P_i, T_i) = G_j(P_j, T_j) \quad (2)$$

$$\text{where } P_i = P_j = P$$

$$\text{and } T_i = T_j = T$$

$$G_i(P, T) = G_j(P, T) \quad (3)$$

From eq. (2) one should be able to search Gibbs free energy for a pressure, P , along isotherms to find the relevant phase boundary. When using SESAME data to create these boundaries one can experience difficulties when attempting to solve the Gibbs intersection for the solid/liquid to vapor transition, due to the Van-der-Waals loops in the underlying expansion models used to create 301 tables in the expansion region of the EOS. For this reason, users are encouraged to employ the legacy “vapor dome” data, which is tabulated in the 401 table. A significant caveat is that when working with tabulated data, one needs to keep in mind that formally equivalent thermodynamic expressions will give different results based on the grid spacing and the selected interpolator. These phase boundaries have been computed using a bi-rational interpolator implemented in EOSPAC6 to maintain consistency with LANL’s use of total EOS tables (301). The 421yyxx tables are tabulated with respect to increasing density, and they define a lower-bound or minimum

temperature boundary for the associated single phase. The 422yyxx tables are also ordered with respect to increasing density, and they represent the upper-bound or maximum temperature for the associated single phase. Together the 421yyxx and 422yyxx tables enclose a single phase region. A best practice includes explicitly tabulating grid points that intersect with tabulated triple-point tables in order to minimize interpolator variance between different SESAME tables.

L. Triple point tables: 423zz series

A triple-point occurs when the Gibbs free energy for three phases are equal at the same temperature and pressure.

$$G_i(P_i, T_i) = G_j(P_j, T_j) = G_k(P_k, T_k) \quad (4)$$

$$\text{where } P_i = P_j = P_k = P$$

$$\text{and } T_i = T_j = T_k = T$$

$$G_i(P, T) = G_j(P, T) = G_k(P, T) \quad (5)$$

Using this data for interpolation can be tricky, since for a (P, T) domain, the Gibb's free energy is represented as a single point in phase space. Alternatively, for a (ρ, T) independent domain, the Gibb's free energy is represented by a line. Only in a (ρ, U) domain can all thermodynamic variables be resolved when provided only two independent variables.

V. SESAME FILE FORMATS

The SESAME data library contains both thermodynamic (e.g., equation of state) and transport coefficients (e.g., opacity and conductivity). Note, for simplicity, the term EOS (equation of state) used herein includes both thermodynamic variables and transport coefficients. With the addition of the new phase specific tables, boundary tables and triple-point tables to the formal definition of SESAME, we were forced to stop using the legacy fixed-format ASCII format[23]. A new ASCII2 format has been defined [24], which closely resembles the legacy ASCII format while relaxing some of the overly-restrictive FORTRAN fixed-format write statements, enabling expanded material identification numbers, implementing expanded table identification numbering, and outlining a more parser friendly whitespace delimited format for the numerical data tables. Reworking the entire format for human readability was considered, but we opted to retain the various data structures so that the ASCII2 format commutes with tables stored the SESAME binary format [4]. Keeping data structures consistent should also minimize the work required for external codes that rely on the ASCII format to update their SESAME parsers to handle ASCII2 files.

Some limited support for conversion between the new ASCII2 format and some of LANL's legacy formats is provided by a utility called `convert_sesame_data`, which is included within the EOSPAC6[5] version 6.5.11 and newer distributions. Previous versions of EOSPAC6 provided limited – albeit incomplete – support as this new ASCII2 format evolved into the current description. New materials contain tables that are **incompatible** with older SESAME formats, namely the original ASCII format[23] and the Helios format,[25]. For the foreseeable future, LANL will only be distributing ASCII2 formatted SESAME data outside of the USDOE[26] complex and both ASCII2 and binary formats internally. Updating codes to be consistent with the new format will be required for modern SESAME materials. We strongly recommend interacting with the data using either EOSPAC6[5] or its officially-supported successor. EOSPAC6 not only provides interpolation and inversion[27] of data, it also provides access to and interpolation of thermodynamically-derived data that is not explicitly tabulated in SESAME.

VI. INTERACTING WITH SESAME DATA

It is tempting to fall into the trap of thinking that SESAME is just a stream of numbers. This assumption might work fine when only converting between formats of SESAME files. Years of combined programmatic experience working with this data would indicate otherwise [28–36]. We recommend interacting with the data using either EOSPAC6 or its officially-supported successor [5].

For the latest information on available SESAME materials, downloading EOS software, and code user manuals can be found on the EOS data team’s internal website [37]. A subset of this information for our external users can be found on our external website [38].

VII. CONCLUSIONS

This document discusses the “state of the art” and foreseeable future of SESAME data produced and published by Los Alamos National Laboratory.

VIII. CONTACT

Any questions concerning this content of this document or the data it describes, should be submitted to sesame@lanl.gov.

IX. ACKNOWLEDGMENTS

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- of the majority of shear data lead us to add this last array as a optional array to the 431 table. We have tried to eliminate any 432 table from the production SESAME database. However any parser written to read SESAME data should consider providing for the optional existence of this table. We will not propagate the 432 table to a phase dependant table construct.
- [9] The original definition of the 431 style tables did not include this Shear modulus at melt array. Most legacy tables in the SESAME database do not have Shear modulus at melt array, any parser written to read SESAME data should provide for the optional existence of this array. The definition of the 431 was expanded to combine the only unique data array from the 432 table. This new definition makes the 432 table obsolete.
 - [10] This table type exists for targeted development purposes inside SESIO[39] and will not be described in this report.
 - [11] The standard definition of the 301-306 and 311 tables includes tabulation of the Helmholtz's free-energy. All new tables should conform to this standard. Most legacy tables in the SESAME database do not have Helmholtz's (old standard), any parser written to read SESAME data should provide for this exception.
 - [12] Los Alamos National Laboratory.
 - [13] All instances of the slash '/' character in the classification marking are substituted with the colon ':' character to avoid conflict with the 101-style comment table delimiter, which is the slash '/' character.
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 - [25] The Helios format is a single-precision variant of the original ASCII format[23] which has been historically used to distribute SESAME data beyond the confines of Los Alamos National Laboratory.
 - [26] United States Department of Energy.
 - [27] Table inversion is a numerical algorithm to change independent and dependent variables of the tabulated SESAME data.
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